

Global usability of mobile CT units in stroke diagnosis: A literature review

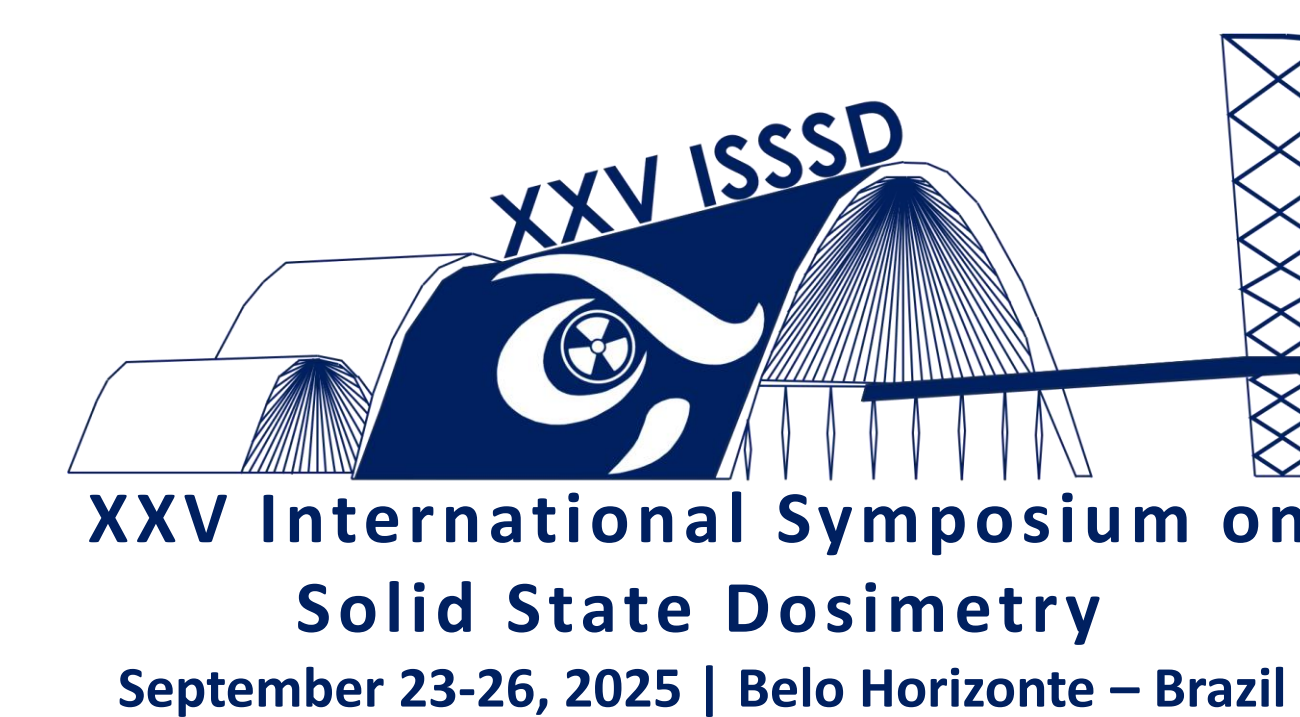
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INTRODUCTION

Stroke is one of the leading causes of death and disability worldwide, with low- and middle-income countries carrying the heaviest burden. In Brazil, it ranked among the top causes of mortality, accounting for more than 99,000 deaths in 2020.

Despite advances in acute stroke care, **timely diagnosis remains a major barrier**, especially in underserved regions where access to imaging is limited. Rapid neuroimaging is essential to distinguish ischemic from hemorrhagic stroke, guide treatment, and improve outcomes.

Computed tomography (CT) is the gold standard, but only **25% of Brazilian public hospitals** offer 24/7 CT access. In many areas, long travel times to imaging centers exceed therapeutic windows, compromising eligibility for thrombolysis.

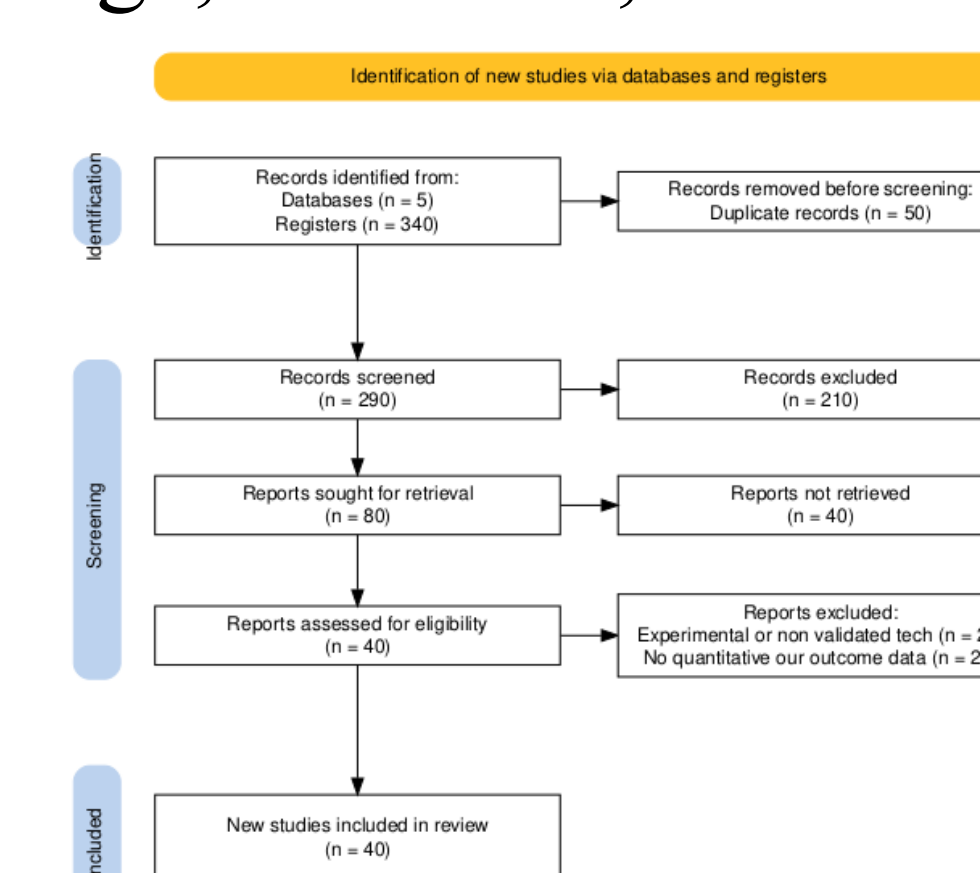
MATERIALS AND METHODS

A **literature review** was conducted on stroke imaging and the global use of Mobile Stroke Units (MSUs) and mobile CT scanners. **Peer-reviewed articles published between 2020 and January 2025** were searched using keywords: “Portable CT”, “Mobile CT”, “Mobile Stroke Unit”.

Inclusion criteria: high-impact journals, studies on implementation and clinical impact, and reports on geographical distribution. Exclusion criteria: non-validated technologies and studies without quantitative data.

A total of **40 articles were selected**, categorized by journal, country, date, and findings. Data were synthesized into graphs and tables following PRISMA guidelines.

Flowchart 01 illustrating the methodology of the review (Haddaway, Page, Pritchard, & McGuinness, 2022).



RESULTS

Publications: 40 articles selected (2020–2025), highlighting robust global evidence. **Most studies originated from the US and UK**; no publications from Latin America.

Trends: Number of **studies increased notably in 2024**, reflecting growing interest and adoption worldwide.

Figure 01: Geographic Distribution of Publications

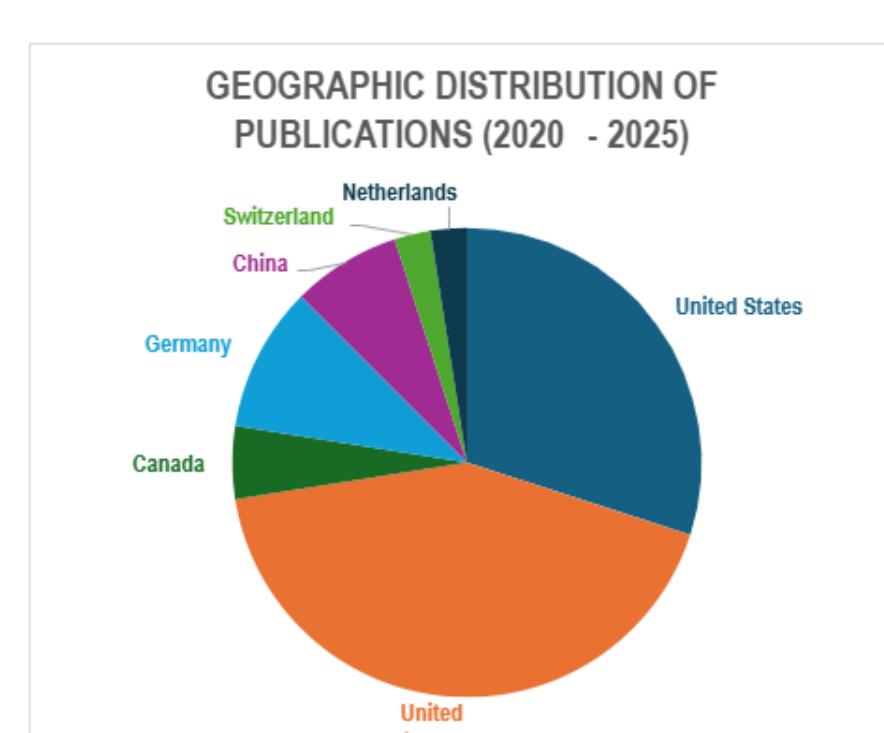


Figure 02: Temporal Trends in Publications



Adoption: Strong concentration in Europe and North America (Germany, Denmark, US). No peer-reviewed data from Brazil/Latin America, though at least one MSU exists in Brasília.

Clinical impact:

- ✓ Higher thrombolysis rates with MSUs (49.9% vs. 9.4%, JAHA 2025).
- ✓ Lower disability at 3 months (JAHA 2021).
- ✓ Reduced door-to-puncture times (94.5 → 41 min, Stroke 2020).
- ✓ Functional independence at 90 days: 55% with MSUs vs. 44.4% standard care (NEJM 2021).

Barriers: High costs, complex logistics, need for workforce training, and limited evidence from low- and middle-income countries.

CONCLUSIONS

Mobile CT units shorten time to treatment and improve stroke outcomes, yet adoption remains uneven, with **significant gaps in Latin America**. This technology represents a **paradigm shift, decentralizing imaging and bringing diagnosis closer to patients, a key step toward reducing global inequities in stroke care**. Broader implementation requires supportive public policies, sustainable funding models, and continuous technological innovation (miniaturization, cost reduction, flexible deployment). Solutions such as Siemens Healthineers' SOMATOM On Site demonstrate that portable CT can achieve fixed-unit quality in ICUs and mobile platforms, proving feasibility for scalable adoption.